

TimeSpan vs. Timefold

Feature Parity, Gap Analysis & Enhancement Roadmap

Executive summary

TimeSpan now ships a genuine, working platform: a custom optimization solver, a live shift-scheduling product, a real multi-tenant admin system, and every one of the platform-level differentiators Timefold markets under its Flow tier (multi-user orgs, config profiles, constraint-level score analysis, roadmap voting, and a support portal), plus most of what Horizon promises (what-if comparison, efficiency X-Ray, benchmarking).

The gaps that remain fall into three distinct buckets, each with a different fix: (1) three of Timefold's four product verticals — field service routing, pickup & delivery, and task scheduling — simply haven't been built yet; (2) a handful of software features are genuinely unbuilt (Copilot, a user-facing custom constraint editor, published API docs/SDKs); and (3) several "Horizon" line items on Timefold's pricing page are service commitments (white-glove onboarding, a dedicated Slack channel, 24/7 support) rather than code, and require an operating process, not an engineering sprint.

This document scores every line item on Timefold's live pricing page against what is actually deployed at timespan.online today, then closes with a set of enhancements that go beyond Timefold's own feature set.

Scoring key

Built — live in production and working end-to-end. **Partial** — a real, working version exists but doesn't yet match the full scope of Timefold's feature. **Missing** — not built. **N/A** — a deliberate product-scope difference rather than a gap.

Spark tier — API-first foundation

Feature	Timefold tier	TimeSpan status	Notes
Pure API access — built for developers	Spark	Partial	The /api/solve endpoint exists and is fully functional, but it isn't published as a versioned, standalone API product with client SDKs the way Timefold sells Spark.
No UI; plug directly into your system	Spark	N/A	TimeSpan is built UI-first by design. A pure headless mode is a possible future SKU, not a gap in the current product vision.
Enterprise solver power / production-grade optimization engine	Spark	Partial	Custom TypeScript constraint solver (solver.ts) with configurable weights and hard/soft constraints.

Feature	Timefold tier	TimeSpan status	Notes
			Real, but a lighter heuristic vs. Timefold's mature CP/local-search solver refined over years.
Maps service integration — travel time & distance matrix	Spark	Missing	No routing/maps engine at all. Directly tied to the missing Field Service / Pickup & Delivery verticals below.
Webhook configuration	Spark	Built	webhooks table + best-effort POST fired on schedule.solved, configurable per user.
1 seat, 1 config profile	Spark	Built	Exceeded: config_profiles table supports unlimited named profiles even on the base tier as shipped.
Quick-start onboarding guides	Spark	Missing	No in-product tutorials or step-by-step first-run guide yet.
Technical API documentation	Spark	Missing	No public API reference / SDK docs page exists yet.

Spark tier score: 2 Built · 2 Partial · 3 Missing · 1 N/A (of 8 tracked items)

Flow tier — team collaboration & visibility

Feature	Timefold tier	TimeSpan status	Notes
Multi-user collaboration	Flow	Built	organizations + organization_members with owner/admin/member roles and RLS.
Unlimited config profiles	Flow	Built	Same config_profiles system, no artificial seat/profile cap enforced anywhere.
Dataset overview — list, search, categorize runs	Flow	Partial	Admin overview and dashboard list recent solve runs; no dedicated search/filter/tag UI for browsing historical datasets yet.
Score analysis & constraint UI	Flow	Built	Constraint X-Ray: structured H1–H4 (hard) and S1–S3 (soft) violation counts persisted per run.
Score graphs & metric relations	Flow	Partial	Coverage / fairness / utilization metrics are

Feature	Timefold tier	TimeSpan status	Notes
			computed and shown as numbers; no time-series graphing yet.
Plan visualizations (schedules & routes on map/timeline)	Flow	Partial	Weekly grid/roster view built; map-based visualization not applicable without a routing vertical.
Metrics & KPI dashboard	Flow	Built	Dashboard overview page: coverage %, feasibility, score, plan distribution.
Product roadmap voting	Flow	Built	roadmap_requests + roadmap_votes with customer voting UI and admin management.
Standard onboarding package	Flow	Missing	This is a service/process line item at Timefold, not software — would need an internal customer-success playbook.
Support portal	Flow	Built	support_tickets table with customer submission form and admin triage view.

Flow tier score: 6 Built · 3 Partial · 1 Missing (of 10 tracked items)

Horizon tier — strategic intelligence suite

Feature	Timefold tier	TimeSpan status	Notes
Efficiency X-Ray — bottleneck detection	Horizon	Built	Same structured breakdown doubles as bottleneck detection (e.g. skill mismatches, overlaps, fairness deviation counts).
Reality checks — real-world feasibility validation	Horizon	Missing	No separate feasibility-vs-reality validation layer distinct from the constraint solve itself.
What-if scenarios	Horizon	Built	Scheduler page solves twice against different weight presets and renders results side by side.
Goal alignment to business objectives	Horizon	Partial	Solve weights let you tune priorities (fairness vs. coverage vs. overtime), but there's no formal UI mapping weights to named business goals.
Comparison UI & benchmarking	Horizon	Built	Same what-if compare view

Feature	Timefold tier	TimeSpan status	Notes
			provides side-by-side score/metric benchmarking.
Copilot — AI-assisted planning recommendations	Horizon	Missing	No LLM-backed assistant layer yet. Highest-leverage gap for investor-facing differentiation.
Real-time traffic (included as standard)	Horizon	Missing	N/A without a routing vertical — see Field Service / Pickup & Delivery gap below.
Staggered release / controlled rollout	Horizon	Missing	No environment/rollout management tooling (single production deployment today).
Custom constraints & model extensions	Horizon	Missing	Constraints are hardcoded in solver.ts; no user-facing constraint builder or plugin model yet.
White-glove onboarding	Horizon	Missing	Service/process, not software.
Dedicated Slack channel	Horizon	Missing	Service/process, not software.
Dedicated optimization advisor	Horizon	Missing	Service/process, not software.
24/7 support	Horizon	Missing	Support portal exists and works end-to-end, but there's no staffed round-the-clock coverage behind it yet.

Horizon tier score: **3 Built** · **1 Partial** · **9 Missing** (of 13 tracked items)

Product verticals

Timefold sells the same platform across four scheduling/routing products. TimeSpan currently ships one of the four.

Feature	Timefold tier	TimeSpan status	Notes
Employee Shift Scheduling API	All tiers	Built	TimeSpan's current and only live product vertical — fully working demo with real solver, roster UI, and constraint breakdown.
Field Service Routing API	All tiers	Missing	Not started. Would require a genuine vehicle-routing solver, not just the shift-assignment engine.
Pickup & Delivery Routing API	All tiers	Missing	Not started. Shares

Feature	Timefold tier	TimeSpan status	Notes
			infrastructure needs with Field Service Routing (maps, distance matrix, capacity constraints).
Task Scheduling API	All tiers	Missing	Not started. Closest to the existing shift-scheduling engine conceptually — the fastest of the three to bootstrap.

Vertical coverage: **1 Built** · **0 Partial** · **3 Missing** (of 4 tracked items)

Hosting & infrastructure

Feature	Timefold tier	TimeSpan status	Notes
Threads / Memory / Tenants published per tier	All tiers	Partial	Plans table stores threads/memory_gb/tenants and renders them on the public pricing page — informational, but not yet wired to real infrastructure isolation or quotas.
Self-hosted deployment option	Horizon	Missing	TimeSpan is managed-cloud-only today; no on-prem/self-hosted packaging exists.

Where TimeSpan goes beyond Timefold

A few things shipped this build that Timefold either doesn't expose publicly or gates behind sales conversations:

- Fully transparent, self-serve pricing — all three tiers and every feature flag are visible and editable in real time from the admin Plans page, and instantly reflected on the public /pricing page. Timefold's Enterprise-equivalent tier is priced as "Custom," pushing prospects into a sales call; TimeSpan's admin can reprice any tier live.
- A real, working admin backend, not a marketing claim — user role management, organization management, plan editing, support ticket triage, and roadmap management are all backed by actual database tables and row-level security, operable today by washington@afriaprocur.com.
- Structured, legible constraint X-Ray — every solve returns named hard/soft constraint codes (H1 skill mismatch, H2 unavailable day, H3 overlap, H4 over-max-shift, S1 fairness deviation, S2 unassigned, S3 back-to-back) rather than a single opaque score, which is easier for a non-technical stakeholder to audit.
- Support and roadmap loops are customer-facing and live simultaneously — a signed-in user can file a ticket and vote on the roadmap from their own dashboard the same day admin can manage both, with no separate CRM or tooling required.
- Auth is already multi-provider — email/password and Google OAuth both work end-to-end in production, which is typically a later-stage integration for early-stage platforms.

Priority roadmap to close remaining gaps

Ordered by investor-visible impact per unit of engineering effort:

- 1. Copilot (AI-assisted recommendations) — the single highest-leverage Horizon gap; even a narrow first version (e.g. "explain this schedule's biggest constraint violation and suggest a fix") demonstrates the intelligence-suite story Timefold sells.
- 2. Task Scheduling API — the second product vertical, and the cheapest to build since it reuses most of the existing shift-scheduling solver rather than requiring a new routing engine.
- 3. Published API documentation + a thin client SDK — required to credibly sell the Spark (headless/API-first) tier at all.
- 4. Real billing (Stripe) — the Plans page currently edits price display copy only; no payment collection or metering is wired up yet.
- 5. User-facing custom constraints — let a customer admin add their own hard/soft rules without an engineer editing solver.ts.
- 6. Field Service Routing + Pickup & Delivery APIs — the largest lift; both need a true vehicle-routing engine and a maps/distance-matrix integration TimeSpan doesn't have today.
- 7. Score graphing, dataset search/filter, and self-hosted packaging — polish items that round out Flow/Horizon parity once the items above are in place.
- 8. Service commitments (white-glove onboarding, dedicated Slack, 24/7 support, dedicated advisor) — these are staffing and process decisions to make once paying customers exist, not code to write now.

Bottom line

TimeSpan already matches or exceeds Timefold on every Flow-tier collaboration and visibility feature, and covers roughly two-thirds of Horizon's intelligence-suite promise. The remaining gap is concentrated, not diffuse: one AI feature (Copilot), one payments integration (Stripe), one documentation effort (public API docs), and three unbuilt product verticals that would each be a multi-week solver-engineering project in their own right. That is a credible, fundable roadmap rather than a rebuild.